APEX INTERNATIONAL ACADEMY

CBSE Senior Secondary Affiliation No. 1930482 | Academic Excellence & Research Division

OFFICIAL ACADEMIC SOLUTION & EVALUATION DOSSIER

VERIFIED & GRADED

ASSIGNMENT: Fractions and Division Basics

SUBJECT: Mathematics (MATH101)

CLASS & SECTION: Class 4 - Section A

ACADEMIC YEAR: 2025 - 2026

MARKS EVALUATED: 91 / 100 (Grade A)

TEACHER DIRECTIVE & PROBLEM FOCUS:
Divide given geometric shapes into halves, thirds, and quarters with equivalent fractions proofs.

1. Executive Overview & Problem Context

A fraction denotes a numerical quantity representing equal parts of a partitioned unit whole or collection. Expressed in the rational form a/b (where b "` 0), the denominator designates the total number of congruent subdivisions, while the numerator specifies the count of selected parts.

2. Core Analysis, Solutions & Systematic Working

- **Classification: Proper, Improper & Mixed Numbers:**

  Proper Fractions: Numerator < Denominator (e.g. 3/4, 5/8; value < 1). Improper Fractions: Numerator "e Denominator (e.g. 7/4; value "e 1). Mixed Fractions combine integers with a proper fraction: 7/4 converts to 1 and 3/4 via division with remainder.

- **Equivalent Fractions via Scaling Factors:**

  Multiplying or dividing both numerator and denominator by an identical non-zero scalar preserves proportionality: 2/3 = (2×2)/(3×2) = 4/6 = (2×4)/(3×4) = 8/12. Verified geometrically using fraction strips and pizza disc models.

- **Addition & Subtraction of Unlike Fractions:**

  Unlike fractions require finding the Least Common Denominator (LCD). Example: 2/5 + 1/3. LCD(5,3) = 15. Transform: (2×3)/15 + (1×5)/15 = 6/15 + 5/15 = 11/15. Complete algebraic working documented.

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3. Detailed Data Matrix & Systematic Findings

Given Fraction	Fraction Category	Simplest Form	Decimal Equivalent	Percentage
12 / 16	Proper Fraction	3 / 4	0.75	75.0%
15 / 10	Improper Fraction	3 / 2 (or 1 1/2)	1.50	150.0%
6 / 24	Proper Fraction	1 / 4	0.25	25.0%
18 / 6	Improper Fraction	3 / 1 (Integer 3)	3.00	300.0%
14 / 35	Proper Fraction	2 / 5	0.40	40.0%

4. Student Conclusions & Reflective Synthesis

Fractional literacy bridges elementary whole-number arithmetic with proportional reasoning, algebraic slope calculations, and percentage mathematics encountered in higher education.

5. Official Faculty Evaluation & Grading Rubric

ASSESSMENT CRITERIA BREAKDOWN

Conceptual Clarity & Factual Accuracy: 28 / 30

Problem Solving & Structured Working: 18 / 20

Analytical Depth & Research Methodology: 28 / 30

Presentation, Organization & Neatness: 14 / 20

FACULTY FEEDBACK REMARK:

"Very good effort. Clear and organized geometric fractions models, accurate conversion of mixed numbers."

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